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Introduction

One of the Grand Challenges for engineering as noted by *The National Academy of Engineering* (NAE) is providing access to clean water. The Academy opens up the discussion of this issue to the American people with the following quote by Samuel Taylor Coleridge "'water, water, everywhere, nor any drop to drink,'" and note that "he did not have the 21st century's global water situation in mind" (National Academy of Engineering, 2008). The NAE assembled a committee to address the challenges that engineers will face in the 21st century. They presented their findings to global leaders throughout the world in order to help eradicate alleviate global problems. One of the key issues that arose was the use of and access to fresh water. "Today, the availability of water for drinking and other uses is a critical problem in many areas of the world" (National Academy of Engineering, 2008). The following pages will examine how governments, non-governmental organizations and businesses can help and promote access to clean drinking water in various regions of the world and why this issue should be a top priority for future global leaders.

Potable and access to sanitary water is a major issue that is often overlooked throughout the world. Clean drinkable tap water and water fountains are commonplace throughout developed countries like the United States. However there are various parts of the world that still struggle with such basic amenities as access to clean fresh water. Access to fresh clean water is of extreme importance to all of humanity. The human body is comprised of upwards of 60 percent of water (The U.S. Geology Survey, 2013). Without water humans cannot survive. In many areas and regions of the world access to such water is very limited, expensive, or non-existent and where there may be access to clean fresh water the water may be contaminated.

"Lack of clean water is responsible for more deaths in the world than war" states The NAE and "about 1 out of every 6 people living today do not have adequate access to water, and more than double that number lack basic sanitation, for which water is needed" (National Academy of Engineering, 2008). This is an important fact to take into consideration. Different regions of the world are perpetually in a state of war. The world's media outlets show the world the harsh reality of war but neglect to show some of the key issues the world continues to face on a daily basis. As noted above, more people are dying because of the simple lack of access to clean fresh water. The leaders of the world need to also consider the fact that their acts of aggression towards other countries and people are not the only source of pain and suffering in the world. The lack of leadership in addressing water issues is of paramount importance because it affects more people than global leaders may realize.

The Issue of Peak Water

As the world becomes more and more populous the need for water becomes ever greater. The idea of 'peak oil', that the world has reached its maximum production of oil, is a term that has become wide spread in economic, political, and environmental debates. Now academics, politicians, and economists alike are considering new 'peak' levels of outputs for other goods. One of those is "peak water". In a book entitled *The Worlds Water 2008-2009: The Biennial Report on Freshwater Resources* Meena Palaniappan and Peter H. Gleick note that "there has been a growing discussion of whether we are also approaching a comparable point of 'peak water,' at which we run up against natural limits to availability or human use of freshwater" (2009).

Not only has the world hit a peak level of oil production and consumption, the world may have also hit a level of peak freshwater consumption, argue the authors. “The finite supply of freshwater on Earth has been nearly tapped dry, leading to a natural resource calamity on par with, or even worse than, running out of accessible, affordable oil” (Palaniappan & Gleick, 2009). They continue by noting “about one-third of the world’s population lives in countries with moderate-to-high water stress, defined by the United Nations to be water consumption that exceeds 10 percent of renewable fresh water resources” (Palaniappan & Gleick, 2009). According to their calculations that means that there were roughly 80 countries, or about 40 percent of the world’s population “suffering from water shortages by the mid-1990s” (Palaniappan & Gleick, 2009).

This is an important issue because of a variety of factors. First, the idea that water has become a commodity in today’s world is troubling. Water is a necessity to life. Water should only be bought and sold to those who can afford to do so. Those who cannot should have readily available access to fresh clean water. Second, water should not be a hot political issue, as it tends to be. In order to sustain our earth and ecosystems global leaders need to find way to provide water to those in need. This will allow for further growth and development of humanity. Furthermore there is a clear need to find an innovative way to provide water to everyone. Global leaders of all walks of life need to prioritize critical issues such as water in order to continue the trend in overall world growth and development.

Oil is a product that can be replaced and substituted with other types of energy in the long run. The world has seen a shift towards renewable and sustainable types of energy that will allow for further economic growth, progression and innovation. However, freshwater is not one of those goods. Humans can survive without oil to power modern machinery, but

cannot survive without the basic element that fuels humanity and the world's ecosystems. This is why access to waters usage and consumption must be of utmost importance for future global leaders. When humanity and the ecosystems that sustain modern day lifestyles are at risk because of limited or no access to fresh and sanitary water problems greater than oil production and global conflicts are bound to arise. Therefore tomorrow's global leaders need re-consider importance and use of water.

Fortunately there are international non-governmental organizations such as the United Nations (UN) that are pushing for reform in the way water is used and distributed in the 21st century. One of those initiatives are known as the Millennium Development Goals.

The United Nations Global Analysis and Assessment of Sanitation and Drinking Water (UN-GLAAS) Report

The UN first established the Millennium Development Goals (MDGs) to promote and pursue the alleviation of a variety of ailments facing the world in the 21st century. These goals set out to eradicate such pressing issues as extreme poverty. The seventh MDG is ensuring environmental sustainability, which includes creating better infrastructure to access clean water throughout the world. Engineers, politicians, and business leaders must find innovative ways to tackle this issue. The UN releases periodic reports regarding the success and shortcomings of the MDGs.

In a 2012 report entitled *UN-Water Global Analysis and Assessment of Sanitation and Drinking-Water* (UN-GLAAS) the World Health Organization (WHO) analyzes the success and failure of this goal. The report was created "from the recognition that the scarcity of

information on national sanitation and drinking-water–related policies, financing and human resources was a major barrier to progress" (World Health Organization, 2012). As noted above there the 21st century faces new challenges that must be overcome through innovative global leadership on all levels.

Potential Business Alternatives: The SODIS Method

Drinking water is a commodity that tends to be taken for granted in the developed world. In today's society drinking bottled water is seen as a cultural norm, even though in countries like The United States fresh clean water from the tap is fine to drink. There are some in modern society that may complain that they do not like the taste of the water out of their tap and opt to buy it in bottles. Bottled water is also an easier convenience for those who live busy fast paced lives where even the simple act of drinking water becomes a commodity.

However, there are still large portions of people throughout the world that do not have access to fresh clean water. There are a variety of underlying factors as to why this might be the case. The following paragraphs will examine a water purification method that can be used to promote access to clean drinking water for those who may not have easy access to it.

To begin we will look at an initiative by *The Swiss Federal Institute of Aquatic Sciences and Technology*. The institute developed a type of water purification that can be done with the sun. SODIS, which is short for solar disinfection, is a method in which old used water bottles are placed in the sun to be purified. The Institute states "for more than 10 years, it has been our mission to provide people in developing countries with access to clean drinking

water" (SODIS, 2013). There are, however, ways in which businesses can find innovative ways to recycle their products.

The Swiss Federal Institute of Aquatic Sciences and Technology is a good example of an organization that is providing a method that is simple to use for the poorest communities of the world and provides businesses with an opportunity to promote the use of their product. Through the promotion of the SODIS method the institute is "trying to improve the supply of drinking water and reduce infant mortality, and thereby make a direct contribution to the millennium development goals of the UN" (SODIS, 2013). As mentioned above The UN's Millennium Development Goals are an important initiative taken on by global leaders to help fix and reduce some of the major issues facing the global community. Organizations such as *The Swiss Federal Institute of Aquatic Sciences and Technology* help provide examples to the rest of the world on how to address critical issues facing the developing world. They strive to help develop methods to aid communities without sufficient access to drinking water. Now those who used to lack basic services such as purified drinking water are beginning to have access to innovative programs that can help them develop non-traditional methods of water purification.

The SODIS method is interesting because of how simple it is. "The SODIS method is ideal for treating water for drinking in developing countries" this is because "all [that] it requires is sunlight and PET bottles" (SODIS, 2013). PET bottles are simply plastic bottles that are used for goods like bottled drinking water and carbonated beverages. They are often recycled in developed countries to be re-utilized to make different products such as planters for gardens or even windows.

However, in the developing world they do not have the infrastructure or government programs and often times PET bottles end up in landfills. With a method like SODIS these bottles can be reused and re-engineered locally to better the community as a whole. This can be done with education on the usefulness of recycling goods that are used everyday. The SODIS method works by simply filling the bottles with unsanitary water and allowing them to sit in the sun for 6 hours (SODIS, 2013). PET bottles are relatively easy to obtain in developing countries and because of the SODIS method now have a second purpose once they are used. Furthermore The Swiss Federal Institute of Aquatic Sciences and Technology notes that "the method is very simple and its application is safe. It is particularly suitable for treating relatively small quantities of drinking water" (SODIS, 2013). The SODIS method has also gained the backing of organizations like the World Health Organization and The Red Cross (SODIS, 2013).

However methods such as SODIS are not perfect. One constraint may be the access to sunlight in regions of the world that have seasons of heavy rainfall. Another important aspect to note is that SODIS is used on a small scale and may only provide small amounts of drinking water. However SODIS is still an important tool that can be used when the skies are clear and the sun is shining. It is important to realize that even the fact that regions of the developing world have easier access to potable water it is an important step in the right direction. Another constraint that SODIS has is that the water remains murky even after it's purified. This may be disconcerting to some, but as mentioned above remains a viable solution that business and global leaders can promote to help alleviate global issues. SODIS is a great example that global leaders can follow when trying to aid the developing world. Humans will always have the desire to help one another. This allows for a better world for us all.

Research Based Approaches to Understanding the Importance of Access to Fresh Clean Water

Local and national governments can help communities become more educated as to why fresh and clean water should be important with the help of research. Often time's communities do not even know or understand why the water that they drink may be detrimental to their livelihoods. Research can be done in communities to help understand and educate governments about the water that their constituents consume in a variety of ways. Engineers, business leaders, and innovators can aid in this development. The National Academy of Engineers notes that:

> In many instances, political and economic barriers prevent access to water even in areas where it is otherwise available. And in some developing countries, water supplies are contaminated not only by the people discharging toxic contaminants, but also by arsenic and other naturally occurring poisonous pollutants found in groundwater aquifers (National Academy of Engineering, 2008).

Often time's communities do not even realize that they are contaminating their water supplies. In some areas of the world it is commonplace to discard trash into water supplies. The people in these communities may not realize or are uneducated about the source of their own drinking water. This may be a source of indirect contamination and is an issue that can be resolved with education about their own water sources.

One potential solution to providing cleaner and safer drinking water is to allow the local communities to develop their own public water systems. In an article in *The Journal of Environmental Health* specifically looking at contaminated drinking water in the Hailun region of China the "results indicate that building public wells in the rural areas of China is a better way of ensuring drinking water safety" (Zhao, Chen, & Zhang, 2013). The study examined the nitrate and ammonia contamination in Hailun, a rural area in northeast China. Specifically the study looked at the contamination levels in domestic tube wells (DTWs) versus contamination

levels in public water supply wells (PWSW). One reason for this, as noted in the study was that "Of the DTWs, an important observation was that the occurrence of groundwater nitrate contamination was directly related to well tube material with different joint pathways" (Zhao, Chen, & Zhang, 2013). This is clearly an issue of infrastructure development within China and is a problem that can be solved by education. This is another opportunity where global business leaders, government officials, and engineers can aid in the alleviation of access to fresh clean water. The study was conducted by researchers by the Chinese Academy of Sciences and which is the national science academy of natural sciences in China.

The article notes, "A high nitrate concentration in drinking water can cause methemoglobinemia in infants and in the stomach in adults" (Zhao, Chen, & Zhang, 2013). According to The United States National Library of Medicines Online encyclopedia symptoms of methemoglobinemia may include "developmental delay, failure to thrive, intellectual disability [and] seizures". All of these problems are devastating for developing countries, especially in rural areas where access to quality medical care may be limited or even non-existent. Furthermore medical issues can strain a family financially and emotionally. Access to cleaner non-contaminated water can help alleviate the stress on rural families in the developing world and help them overcome the burdens they face. This can be done with practical measures taken by non-governmental, business, and governmental initiatives.

Water Wars: Political Issues

Water can also be a hot political issue. When governments try and divert water from one are to another to help sustain or develop other regions of a given issue political tension can

arise. The NAE notes, “diverting water to some people often means less for others and can become an explosive political issue” (National Academy of Engineering, 2008). Governments face challenges when they attempt to divert water from one region to another. Many industries require water to sustain and develop factories and enhance production.

Specifically developing countries face these types of challenges because they rely may rely heavily on water intensive means of production such as agriculture. Agriculture is important to developing economies because it may be one of the only productive and developed industries that may be present. Furthermore it may be one of the only skill sets that a given citizenry has. Without proper irrigation and fertile land rural areas that rely on agriculture to be economically viable may succumb to natural fluctuations in rainfall and are therefore vulnerable to droughts. Without the correct solutions to the water needs greater political tension and risk may arise. There are a variety of downfalls to a country that is seen as political risky. This is because leaders of business and industry innovators may not want to enter a country because of the political risk.

The importance of developing global and innovative leaders on both the governmental and business level is key for countries to develop and become prosperous. Without proper attention given to key issues such as access to and the use of fresh clean water countries may fall into political turmoil and shun away potential development and growth opportunities.

Conclusion

The World Health Organization will publish a report in 2013 entitled “UN-Water Country Briefs” This pilot project will “provide a strategic outlook on the critical importance

of investments in water for human and economic development at country level" (World Health Organization, 2012). This report will allow organizations like the United Nations to achieve more of the Millennium Development Goals and will allow countries to see how they can better provide clean water for their citizenries. The world will become a better place for all with the push of world leaders to improve access to potable and sanitary water. Without water, sustainable human existence is not possible.

In conclusion it is important for the future global leaders of the world to develop and adapt innovative solutions to global problems. As shown above this can be done through successful education in all aspects of leadership. With the correct business, political and engineering initiatives goals presented by international non-governmental organizations such as the United Nations Millennium Development Goals can be achieved. Without new approaches to the problems that the world faces in the 21st century the youth of today may never see tomorrow.

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